

THE ECHO BELT

京张回响带

Steam Whistle, Bell, Code - Three Echoes · One Spine, Two Wings, Three Zones

Centennial Jing-Zhang AI Innovation Belt Open Call · A3 Proposal Booklet

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Provisional boundary notice: all boundaries are provisional constraints (not official redlines, boundary_precision=provisional_rough); all areas recomputed in EPSG:4548 as low-confidence design-model values; the whole package is recalculated once official data is released. All spatial content is a conceptual recommendation for professional teams to deepen.

Design basis

Primary basis: the pre-qualification announcement (scopes, key areas, tasks, depth). Second: the agent-facing taskbook (positionings, functions, zones/wings, agent.1-agent.6). Structured basis: the repository site package and public source registry. Formal claims use formal-ready sources only; cases and industry context are background.

Overall concept

The Echo Belt: the 1909 whistle marked the origin of China's independent innovation; a century later an AI belt grows along the same corridor. The switchback 人 translates into a human-centred AI city; the structure is one spine (Echo Spine), two wings (Zhongguancun Services x Xiaoyue River Scenarios), three zones (Zhongzhiyuan, Origin Community, Dazhongsi).

Three scope levels

Coordinated Research Area approx. 43.6 sq km: strategy for the AI ecosystem and future city form. Overall Design Area approx. 11.4 sq km: renewal framework and regulatory-plan-level urban design (submitted boundary recomputes to approx. 11.41 sq km, +0.1%). Key areas approx. 368.4 ha: Zhongzhiyuan (192.1), AI Origin Community (104.3), Dazhongsi (72.0).

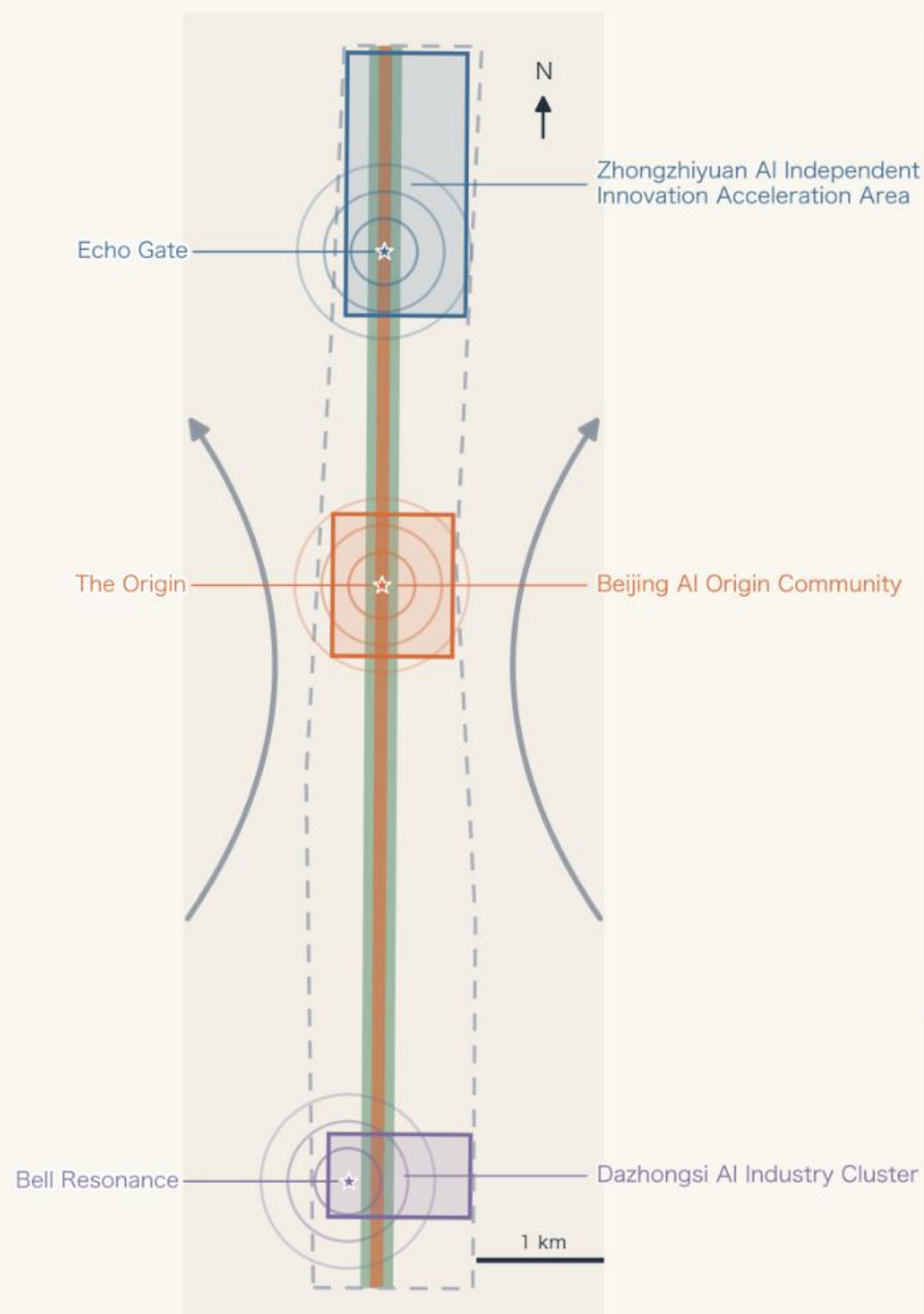
Data boundary

No official precise redline is published: all boundaries are provisional constraints for generation, display, and self-check only; FAR/height controls remain pending; the constraints layer stays honestly empty with a declared gap; the package is fully recalculated when official data arrives.

Overall Concept · One Spine, Two Wings, Three Zones - Three Echoes

Fig.1 / Overall Design Area (approx. 11.4 sq km, provisional boundary) - conceptual recommendation

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Three Echoes - the cultural storyline

- 1909 Steam Whistle
The Jing-Zhang Railway: origin of independently engineered Chinese trunk rail; the switchback was the smartest spatial fix of its era.
- Yongle Bell
The Yongle Bell at Dazhongsi: the city's civic bell, translated into public memory and a sound landmark for the southern cluster.
- AI Code
A century later the same corridor grows an AI innovation belt: an open, verifiable, human-centred echo.

The 'ren' (人) double meaning - spatial structure

Switchback rail (人) → a human-centred AI city.
One spine: the Echo Spine along the Jing-Zhang Railway Heritage Park - a continuous public axis with E-W stitching and N-S continuity;
Two wings: Zhongguancun Technology Services Wing x Xiaoyue River Scenario Wing;
Three zones: Zhongzhiyuan (full-stack autonomy) - AI Origin Community (innovation ecosystem) - Dazhongsi (native-AI programs).

Legend

- Echo Spine green corridor (conceptual)
- Echo Walk public promenade (conceptual)
- Three key areas (provisional extents)
- AI pilgrimage landmarks (3, conceptual)
- Provisional boundary (not an official redline)
- Wing synergy directions (indicative)

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Fig.1 - one spine, two wings, three zones and the three echoes; provisional boundary as low-contrast dashes

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Naming and identity (agent.1)

Master brand The Echo Belt; spatial sub-brands Echo Spine and Echo Walk; event brand Echo Summit; community brand Echo Commons. The logo direction prototypes the character 人 - two strokes at once the switchback and a pair of sound-wave lobes, deep ink blue x signal orange; a design direction only, formal identity to be rights-cleared.

Five-function synergy loop

Sourcing (Origin Community) → acceleration (Zhongzhiyuan) → conversion (Dazhongsi) → services (Zhongguancun Wing) → experience (Xiaoyue River Wing) → back to sourcing. The three positionings map to the narrative system, the Echo Walk with scenario cards, and the three-zone industrial organisation.

Global cases (agent.2, six)

Station F (release mechanism), one-north (themed quarters), Kendall Square (near-campus mixed ground floors), King's Cross (rail-heritage translation), Zhangjiang AI Island (whole-area scenario openness), MaRS (non-profit operating hub). Shared lesson: a stage to release + a field to test + a street to stay - assigned to the Origin Plaza, the testing loop, and the Echo Walk. Background references only.

Factor mechanisms (eight lists)

Land (stock first), space (shared testing/release facilities), industry (three-zone division), capital (conceptual guidance direction), talent (apartments + international community), computing (distributed edge), data (compliant circulation spaces), scenarios (open catalogue). Mechanism suggestions only - no funding, investment, or policy commitment.

Brand Identity and Logo Concept Applications - the character 人 x sound wave

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Fig.7 / Design direction only: no trademark search or registration performed; the formal identity must be separately rights-cleared; no existing typeface, trademark, or corporate mark is used

Construction: two strokes, two wave lobes



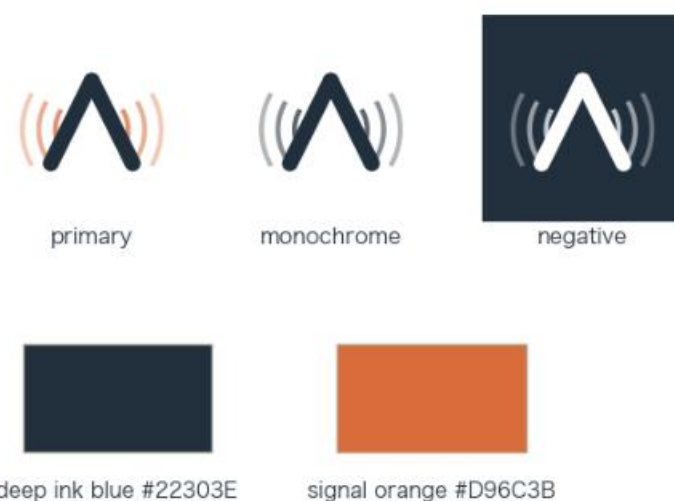
Wayfinding pylon (Echo Walk)



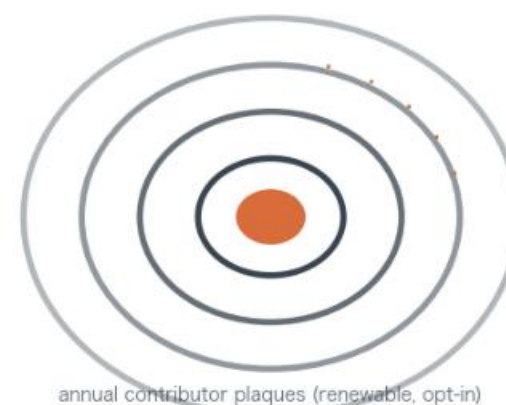
Echo Summit banner



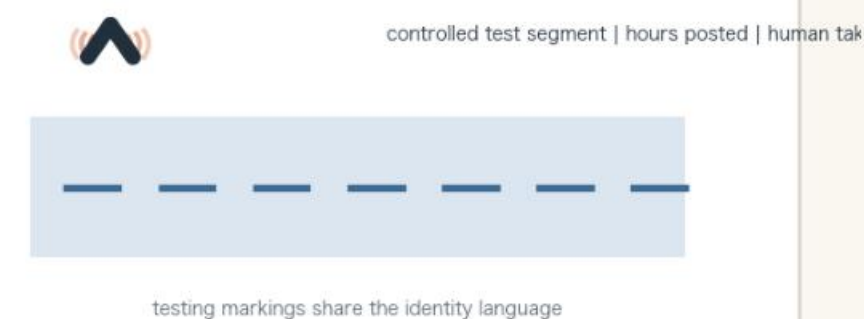
Variants and colours



Honor Ring paving module (Origin Plaza)



Open-testing marking (Zhongzhiyuan loop)



Usage rules: monochrome and negative versions allowed; animatable as a sound wave; wayfinding and testing markings share the waveform-tick language; bilingual
All applications are conceptual mock-ups for professional design teams to deepen; not a final visual identity.

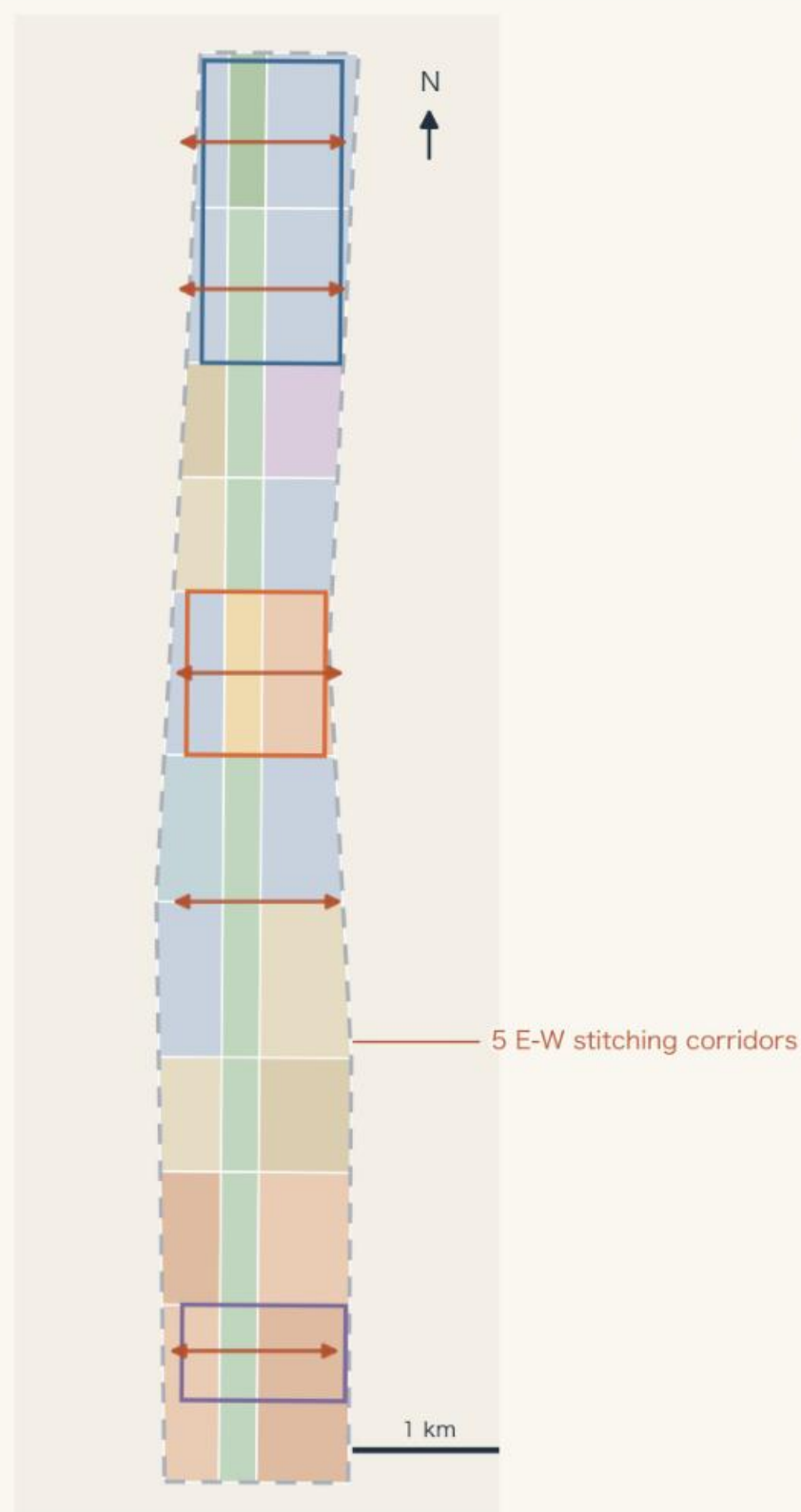
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Fig.7 - logo construction, variants, colours, and four concept applications; no trademark search, formal identity to be rights-cleared

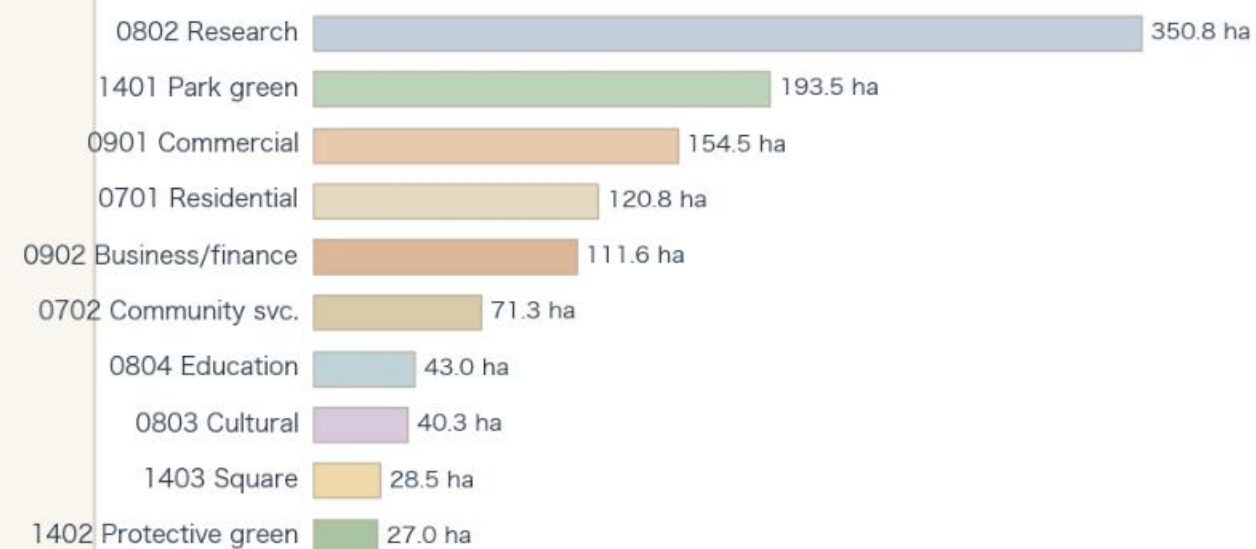
Land-Use Layout and Spatial Structure - a seamless design-model partition

Fig.2 / Land uses are conceptual recommendations for professional deepening; regulatory controls pending official conditions

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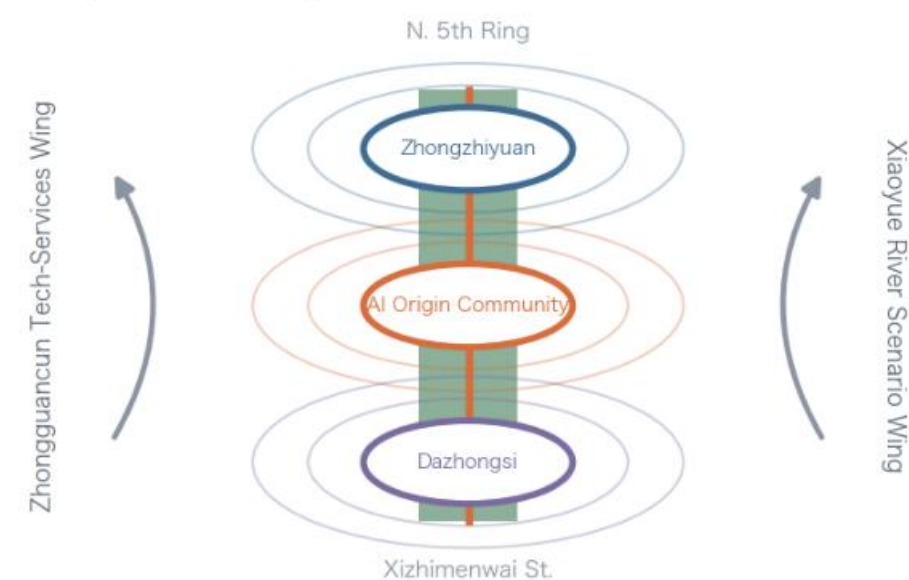


Land-use composition (recomputed in EPSG:4548)



Codes follow the MNR land-use classification guide; shares are design-model values, not statutory controls. Green ratio 21.2%, public-space ratio 11.8%.

Structure: one spine, two wings, three zones



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Fig.2 - ten classes, 30 units, seamless partition (EPSG:4548); conceptual land uses

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Detailed Design of the Three Key Areas - three zones, three colours, three echoes

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Fig.3 / Key-area extents are rough provisional polygons; every spatial move is a conceptual recommendation



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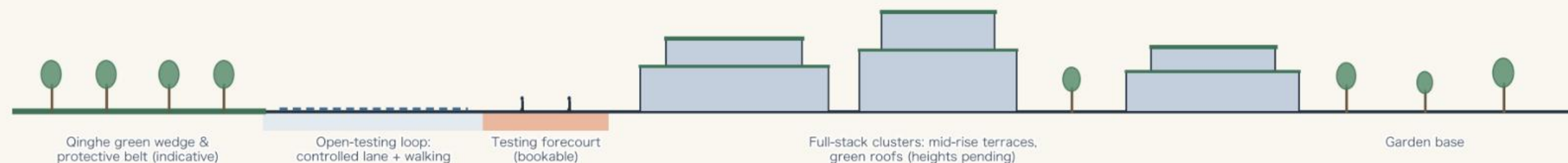
Fig.3 - three zones, three colours, three echoes: Zhongzhiyuan (Echo Gate), Origin Community (The Origin), Dazhongsi (Bell Resonance)

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Differentiated Spatial Prototypes and Conceptual Sections of the Three Key Areas

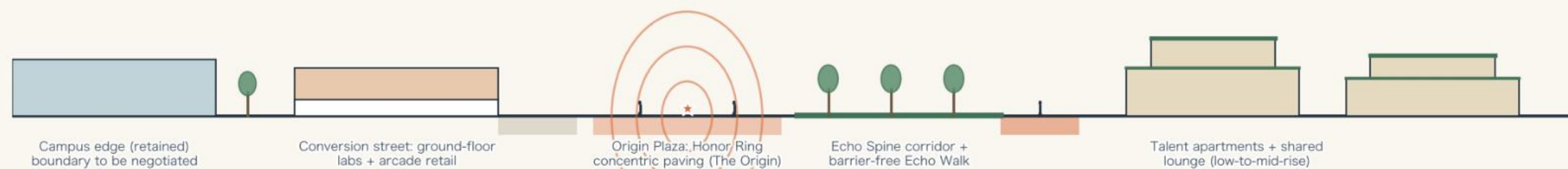
Fig.6 / Conceptual sections are indicative spatial prototypes, not engineering design; dimensions, heights, and redlines await official conditions and professional deepening

① Zhongzhiyuan garden-type full-stack district: green wedge - testing loop - research clusters (conceptual)



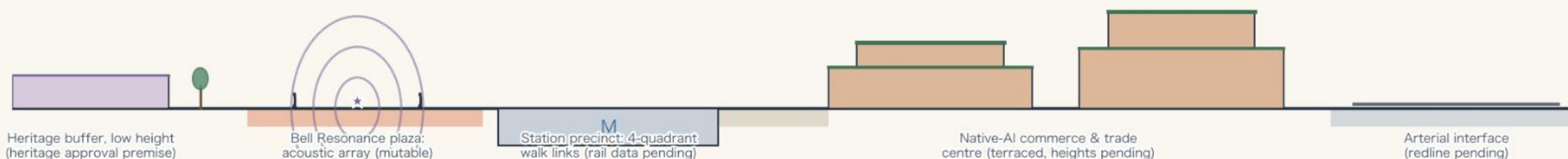
• Prototype: low-density garden base; the open-testing loop is a dual-mode street with controlled switching; Qinghe blue line unpublished - waterfront indicative only

② Origin Community near-campus prototype: campus edge - conversion street - Origin Plaza - Echo Spine (conceptual)



• Prototype: mixed frontage of ground-floor labs + arcade retail; Honor Ring paving organises the plaza in concentric rings; campus boundary and ownership to be negotiated

③ Dazhongsi urban station-city prototype: heritage buffer - Bell plaza - station precinct - commerce (conceptual)



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Fig.6 - garden / near-campus / station-city prototypes; sections are indicative, not engineering design

Regional innovation-synergy matrix

Five counterparts handled separately - Zhongguancun AI Beiwei Community, Future Science City, Huairou Science City, BDA, Beijing-Tianjin-Hebei: factual sentences cite official background sources (the Beijing STIC Regulation and public Beiwei reporting); everything else is explicitly a conceptual mechanism pending negotiation, implying no agreed cooperation. Interfaces: release hall / evaluation centre / trade centre / Echo Summit / open scenario catalogue.

Stage gates for test scenarios (03-06)

The four test scenarios set entry conditions, pause/exit triggers, data-responsibility chains, and review outputs between controlled test, evaluation, and limited opening; numeric thresholds are fixed by authorities and operators in pilot plans - no fabricated precision; testing never implies approved operation.

Implementation-grade matrix EB-01 - EB-08

Each of the eight conceptual projects now carries: suggested responsible-party type, prerequisite materials, pilot deliverables, suggested acceptance indicators, pause/exit triggers, and review outputs; acceptance indicators are suggested conventions, operating-cost sources await dedicated studies, and everything is a suggestion, not an approved arrangement.

Participation and inclusiveness

The council includes resident representatives (suggested quarterly meetings + annual public deliberation); child/elderly trial sessions and accessibility-group trials feed the stage gates; governance AI gets an objection channel with published deadlines; every AI service keeps a no-digital-device alternative. All operational suggestions.

Twelve scenario cards

01 release hall, 02 Honor Ring, 03 autonomous shuttle [TEST], 04 service-robot validation [TEST], 05 model safety sandbox [TEST], 06 sensing & wayfinding [TEST], 07 walking companion, 08 health pre-triage, 09 education open labs, 10 legal & civic guide, 11 native-AI retail & data elements, 12 Bell soundscape. Six elements per card: carrier / group / data / privacy / human review / operator.

Three-stage testing

Controlled test → evaluation → limited opening; a test scenario never implies approved operation; scenarios that cannot be human-reviewed are excluded; no scenario may require non-public data, personal privacy, or a designated vendor.

Six personas

Open-source developers, startups, faculty & students, enterprise R&D and international visitors, residents (incl. elderly and children), city operations staff - each with spatial responses and privacy/review boundaries; no individual profiling, aggregate statistics only.

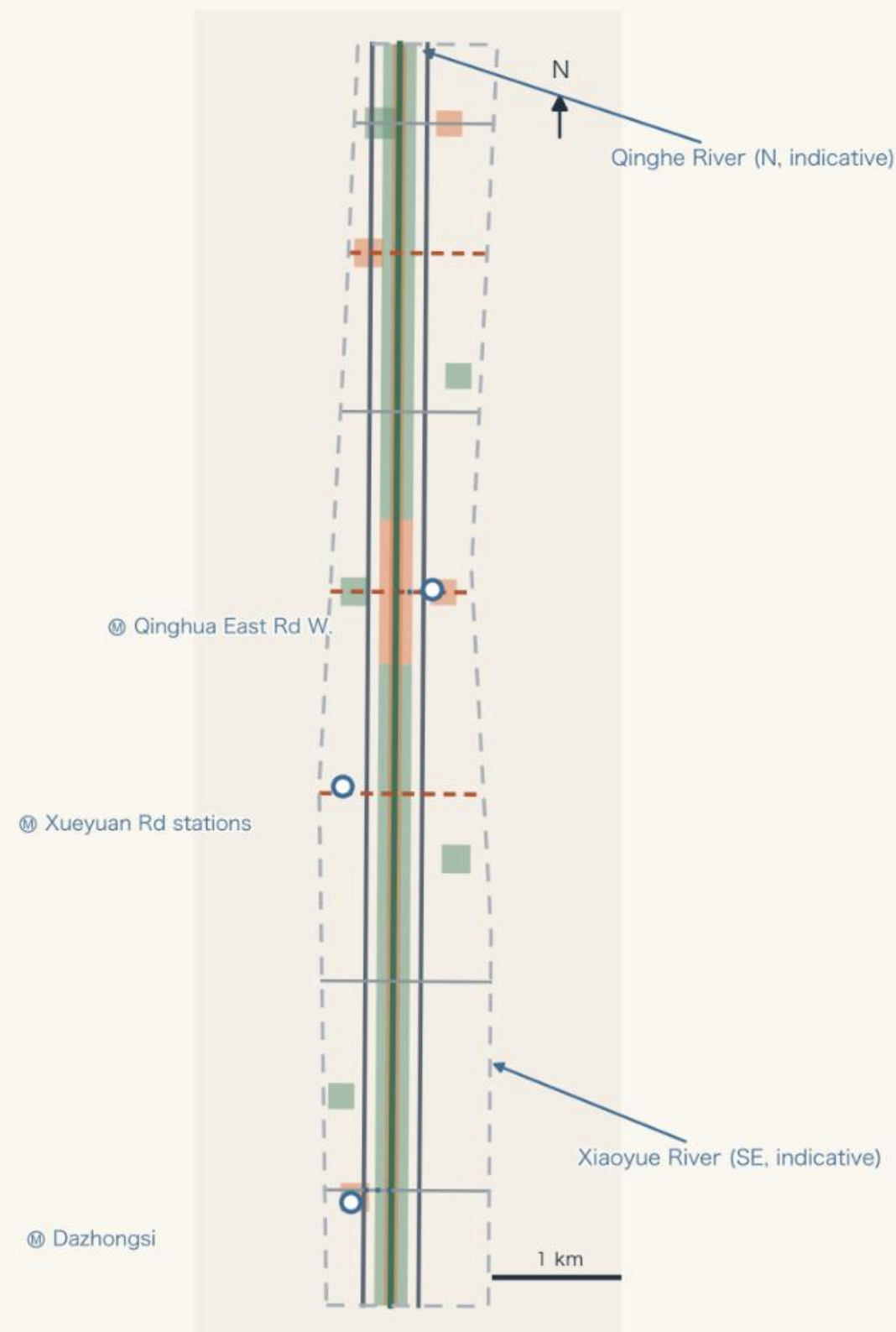
Governance principles

Data minimisation, explainability, switch-off capability, final human review; generative services read through Article 2 of the Interim Measures; public service venues keep human channels per the listed contexts of Article 39 of the Barrier-Free Environment Law.

Mobility, Walking-Cycling, and Blue-Green Public Space System

Fig.4 / Roads are conceptual micro-circulation and slow-mobility suggestions, not redlines; water annotations are indicative

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Legend

- Echo Walk greenway spine (approx. 9.7 km N-S)
- Slow-mobility stitching corridors (E-W)
- Conceptual micro-circulation service streets x2
- Transit connections (indicative stations)
- Green and open space
- Public space / plazas

Slow-mobility and interchange logic (conceptual)

- One spine: the Echo Walk runs barrier-free end to end, linking 3 zones, 2 stations, 3 landmarks;
- Five stitches: E-W slow corridors bridge severances of rail heritage and fast roads;
- Two service streets: micro-circulation keeps the promenade continuous and calm;
- Station-city integration: Dazhongsi and Qinghua East Rd W. forecourts plug into the Walk;
- All alignments are conceptual; redlines and rail alignments follow official data.

Blue-green network (indicative)

Green ratio 21.2% - public-space ratio 11.8% (design-model values, EPSG:4548). North links the Qinghe waterfront corridor, south echoes the Dazhongsi heritage setting, east joins the Xiaoyue River scenario wing; blue lines, flood and ecological controls are unpublished, so all waterfront ideas remain conceptual directions for professional deepening.

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Fig.4 - one spine, five stitches, two service streets; all alignments conceptual

Eight conceptual projects

EB-01 Walk continuity, EB-02 Origin Plaza & Honor Ring, EB-03 conversion street, EB-04 testing loop & evaluation centre, EB-05 Echo Gate & Qinghe wedge, EB-06 Dazhongsi integration & Bell plaza, EB-07 mid-belt interfaces, EB-08 component library - each with type, phase, and dependencies; all reference schemes.

Annual events (agent.6)

Flagship Echo Summit (autumn) + spring Open-Source Season + summer Testing Week + winter Bell Night; the Echo Commons runs as online repository + three anchors (release hall / academy / testing loop) with a community council, the Honor Ring roll as annual ceremony; scenario opening follows catalogue-application-evaluation-exit. All operational suggestions.

Three phases

P1 1-3 yr (approx. 5.15 sq km): Origin Community and mid-north spine; P2 3-5 yr (approx. 3.69 sq km): Dazhongsi station-city integration; P3 5-10 yr (approx. 2.57 sq km): mid-belt stitching and full scenario coverage. Matches the phasing layer.

Communication and conversion

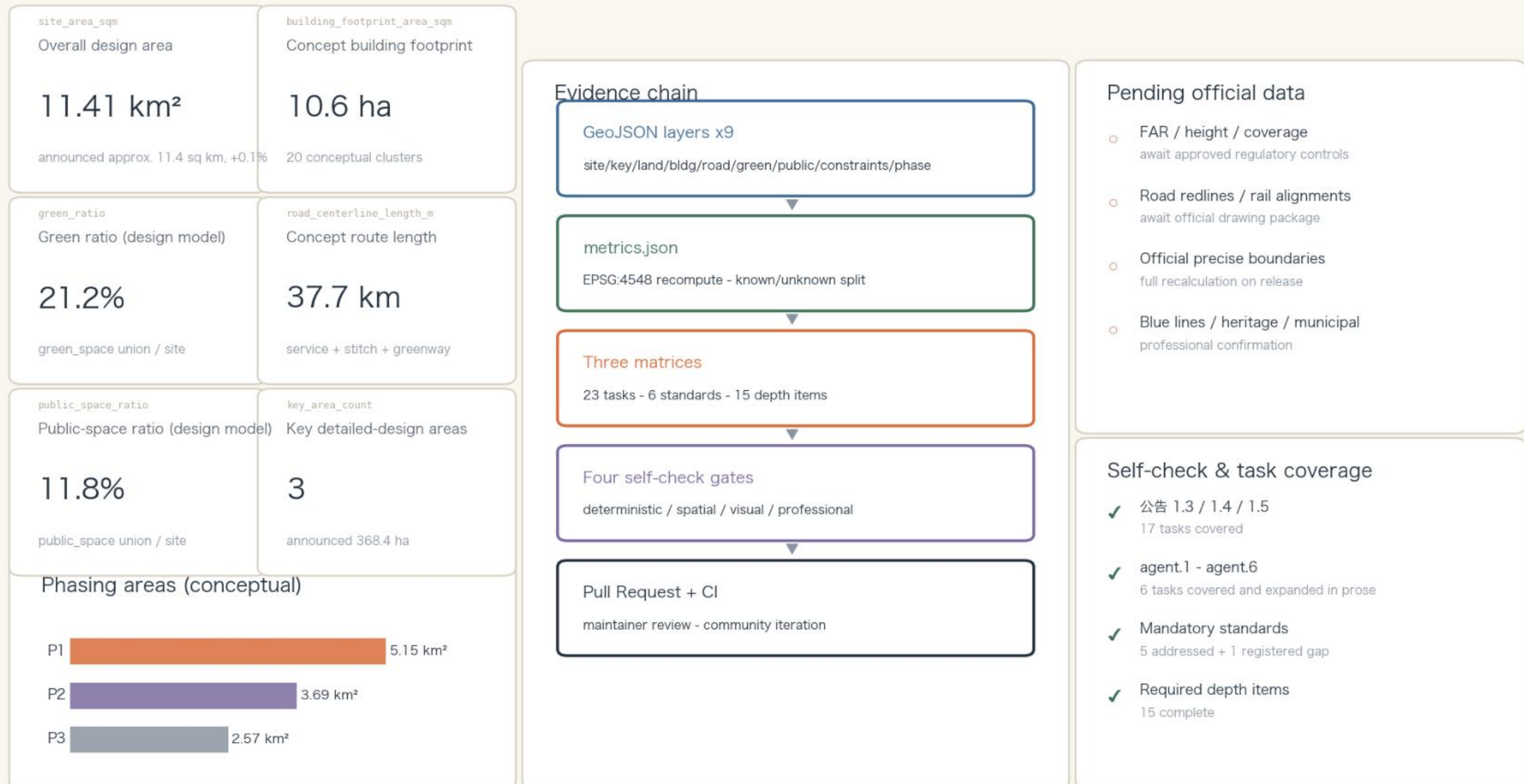
One-line narrative: from the first whistle to the next echo. The Echo Summit is the entrance, converting through the Zhongguancun Services Wing into an attend-test-settle funnel; no promised outcomes.

Metric Recomputation and Evidence Chain - a verifiable design model

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Fig.5 / All known metrics recompute from the submitted GeoJSON in EPSG:4548; control metrics await official conditions

Core metric cards (low-confidence design-model values; recalculated when official data arrives)



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Fig.5 - all known metrics recompute; control metrics pending official data

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Spatial data risk

Provisional boundaries never serve as redlines, approvals, or precise areas; regulatory, redline, blue-line, heritage, and municipal conditions are unpublished, so related conclusions are pending; the package fully recalculates on official release per registered triggers.

Copyright and generation disclosure

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Boundary-clause conformance

No statutory FAR/height/intensity judgments, no parcel demolition conclusions, no road/rail/bridge/pipeline engineering schemes, no ownership/investment/approval judgments, no non-public data, no idea presented as a confirmed arrangement; all spatial suggestions are conceptual recommendations / reference schemes for professional deepening.

Core metrics

Overall area approx. 11.41 sq km; green ratio 21.2%; public-space ratio 11.8%; concept footprint 10.6 ha; concept routes 37.7 km - all low-confidence design-model values recomputed in EPSG:4548.